GLYCEMIC CONTROL



Studies have reported an association between hyperglycemia and adverse infectious and cardiovascular outcomes after cardiac and noncardiac surgery. [1-5]

Many studies recommend delaying elective surgery until until hemoglobin A1c (HbA1c) levels are below 8.5, recognizing that this may not be feasible for all patients and that such a cutoff is not currently supported by robust evidence. [6-10]

Screening Tools

Glycemic Control Screening Questions extrapolated from BC Guidelines for Diabetes Care [11]

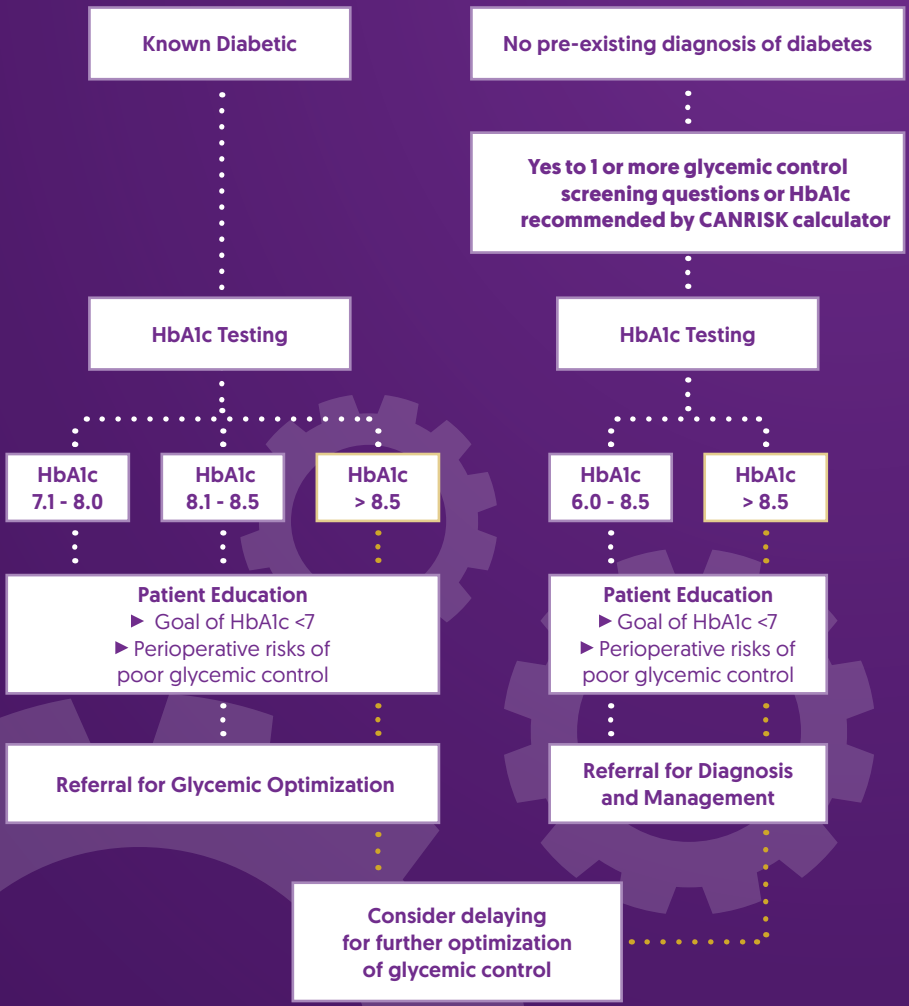
The CANRISK questionnaire is a validated risk tool for patients to self-assess their diabetes risk. [11]

Glycemic Control Screening Questions

	YES	NO
Previously diagnosed with diabetes	☐	☐
Age over 40	☐	☐
Family history of diabetes	☐	☐
History of hypertension, dyslipidemia or steroid use	☐	☐
BMI >25	☐	☐
History of gestational diabetes mellitus	☐	☐
History of pre-diabetes	☐	☐

SCORING If **YES** to any of the above questions then a preoperative screening hemoglobin A1c test is recommended.

GLYCEMIC CONTROL ALGORITHM



Prehabilitation and Optimization Recommendations

Patient Education	⚙ Reinforce general goal of A1c less than 7 [12] ⚙ Explain perioperative risks of hyperglycemia ⚙ Refer to online patient resources ⚙ If no previous diabetes diagnosis: • HbA1c 6.0-6.4: prediabetes likely • HbA1c > 6.5: diabetes likely
Referral for Diabetes Diagnoses and Management (no pre-existing diabetes)	⚙ Encourage follow-up with primary care provider for confirmation of diabetes diagnosis and initiation of management ⚙ Refer to online provider resources for referral templates
Referral for Glycemic Optimization (known diabetes)	⚙ Request re-evaluation from physician directing diabetes care with the goal of improving blood glucose control prior to surgery, if feasible ⚙ GLP-1 receptor agonists may increase aspiration risk due to delayed gastric emptying. Institutional practices vary by center and may change as evidence evolves. Contact your local surgical or anesthesia department for site-specific guidance before advising patients. ⚙ Refer to online provider resources for telehealth consultation options and referral templates
Consider Delaying for Glycemic Optimization	⚙ Discuss concerns, including potential benefits and risk of delaying surgery, with surgical team, particularly for total joint arthroplasty, vascular surgery, and spine surgery with instrumentation ⚙ Refer to local diabetes expert or online provider resources for telehealth consultation options